

Практикум

E1:archive

Every past-paper question on this topic, grouped by technique. Not a practicum — a drill. There is no theory here and no ladder to climb: the theory is in Practicum E1, and this notebook is what you open afterwards, when the only thing left is to do them all until the moves are automatic.

О практикуме

Пятый практикум серии на английском, устроенный как B2–B5: ноутбук, карточка приёмов и сообщения проверок английские, документация репозитория остаётся русской.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-e1-limits

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

E1 archive: limits, indeterminate forms, l'Hôpital's rule

Every past-paper question on this topic, grouped by technique. Not a practicum — a drill. There is no theory here and no ladder to climb: the theory is in *Practicum E1*, and this notebook is what you open afterwards, when the only thing left is to do them all until the moves are automatic.

What is inside. The whole of `calculus.limits`, sessions May 2021 — November 2025: **24 questions, 73 marks**, in nine sections, one section per technique.

The corpus records 22 blocks and 81 marks. Two of them are duplicates. The November 2023 Paper 3 is a single paper filed under Common, but the corpus keeps a zonal copy for each of TZ1 and TZ2 and the two pages are identical byte for byte. And the November 2023

question on $\sin^2(kx)/x^2$ was set in TZ1 and in TZ2 word for word — two papers, one question. Each appears here once. Nothing else is left out and nothing is added.

Eleven questions are Paper 1, five are Paper 2, eight are Paper 3. But the usual split by paper does not apply to this topic: the answers are $\frac{2}{3}$, -3 , $-\frac{1}{4}$, $\frac{\pi}{4}$, $-\frac{n}{2}$, $\frac{1}{2}n(n+1)$ almost everywhere, calculator or not. Only two answers in the whole topic are numbers you would want a calculator for, and both are values of a constant rather than limits.

How to work. Read the question, answer in the cell below it, run the cell. Most of these checks do not know the answer. `verify_limit` takes the expression from the question, walks in towards the point along $10^{-1}, 10^{-2}, \dots, 10^{-8}$ from both sides, and asks whether the number you named is where the values are settling. There is no stored constant for it to be wrong about — so an answer reached by

a wrong route still has to be true, and an answer with x still in it is refused on sight.

Two checks do compare a hash, and they are the same case twice: the answer is not a limit but a constant chosen to make one come out right — $k = 4$ so that $\lim_{x \rightarrow \infty} \frac{\sin^2 kx}{x^2} = 16$, and $k = 9$ so that a density integrates to 1.

One question has no cell at all: 9.1 asks, in words, why a limiting speed is meaningless for a runner in a 200 m race.

Leave a cell blank and it prints  and moves on, which means you can run the whole notebook top to bottom on the first open and nothing breaks.

The solutions are at the end, numbered the same way. Open one after you have worked the question, not before — and read the markscheme note in it, because that is where the marks actually are.

The nine sections.

§	technique	questions	marks
1	Substitute, and cancel what stops you	1	part of 3
2	The highest power at infinity	2	part of 6
3	l'Hôpital's rule, one round	3	8
4	l'Hôpital's rule, more than one	3	18
5	Maclaurin instead	3	9
6	The constant that makes a limit exist	3	10
7	The limit taken in a parameter	3	9
8	The rule with a letter inside the expression	3	9
9	What the limit means	2	part of 5

Sections 1-2 need no calculus at all. Sections 3-5 are the rule and the alternative to it, and they carry 35 of the 73 marks. Section 6 works backwards from a limit to a constant; sections 7 and 8 leave a letter in the answer. Section 9 is four marks for two sentences.

One habit runs through all of it. Write $\lim_{x \rightarrow a}$ on every line. May 2021 caps a perfect solution at four marks out of five without it; May 2024 withholds the final **A1**; November 2022 refuses the mark for the indeterminate form outright.

```
import sys
sys.path.append('..')          # from
practicum/calculus to practicum/kit/
import sympy as sp             # the escape
hatch: anything not in kit is in sp
from kit import *              # checks +
Interval, Rational, oo, pi

language('en')                 # this
notebook is in English, and so are the checks

a, b, c, m, n = symbols('a b c m n')
alpha, F = symbols('alpha F')
# F stands for f(x, y) in section 7: a
gradient that does not move while
# alpha does. Naming it as a letter is what
lets a check substitute values.

print('ready; sympy', sp.__version__)
print('an exact limit:      ', Rational(2,
3))
print('a limit in terms of n:', -n/2)
print('an infinite limit:   ', oo)
print("a form, as a string: ", '0/0')
```

1. Substitute, and cancel what stops you

Substituting is the first move, always. When it gives a number, that is the limit. When it gives $\frac{0}{0}$ because the top and the bottom share a factor, cancel the factor and substitute again.

1.1 — May 2024 TZ1 Paper 1 Q11(f) (i), part of 3 marks

Consider the polynomials $P(x) = 3x^3 + 5x^2 + x - 1$ and $Q(x) = (x + 1)(2x + 1)$, and the function

$$f(x) = \frac{P(x)}{(x + 1)Q(x)}, \quad x \neq -1, x \neq -\frac{1}{2}.$$

Find $\lim_{x \rightarrow -1} f(x)$.

```
FX = (3*x**3 + 5*x**2 + x - 1) / ((x + 1) *  
(x + 1) * (2*x + 1))
```

```
q1_1 = ...
```

```
verify_limit('1.1', q1_1, FX, point=-1)
```

2. The highest power at infinity

Divide the top and the bottom by the highest power of the variable that appears, then use $\frac{1}{x^k} \rightarrow 0$. A root counts as a power: $\sqrt{t^2 + 0.2}$ behaves like t .

2.1 — May 2024 TZ1 Paper 1 Q11(f) (ii), part of 3 marks

With f as in 1.1, find $\lim_{x \rightarrow \infty} f(x)$.

```
q2_1 = ...
```

```
verify_limit('2.1', q2_1, FX, point=oo)
```

2.2 — May 2025 TZ3 Paper 2 Q10(c) (i), part of 3 marks

Two athletes compete in a 200 metres race along a straight track. Fiona's velocity, in ms^{-1} , during the race can be modelled by

$$v(t) = \frac{8.14t}{\sqrt{t^2 + 0.2}}, \quad t \geq 0,$$

where t is measured in seconds from when the race starts.

Write down the limit of $v(t)$ as t approaches infinity.

```
VT = Rational(814, 100)*t / sqrt(t**2 +  
Rational(2, 10))
```

```
q2_2 = ...
```

```
verify_limit('2.2', q2_2, VT, var=t,  
point=oo)
```

3. l'Hôpital's rule, one round

If substituting gives $\frac{0}{0}$ or $\frac{\infty}{\infty}$, differentiate the numerator and the denominator **separately** and substitute again. The rule reads quotients only: a product $\infty \cdot 0$ has to be turned into one first.

3.1 — May 2021 TZ1 Paper 1 Q8, 5 marks

Use l'Hôpital's rule to find $\lim_{x \rightarrow 0} \left(\frac{\arctan 2x}{\tan 3x} \right)$

.

```
q3_1 = ...
```

```
verify_limit('3.1', q3_1, atan(2*x)/tan(3*x))
```

3.2 — May 2023 TZ1 Paper 3 Q1(c)(i), 2 marks

The area of the region bounded by $y = xe^{-x}$, the x -axis and the line $x = b$, where $b > 0$, is $\frac{e^b - b - 1}{e^b}$.

Use l'Hôpital's rule to find $\lim_{b \rightarrow \infty} \frac{e^b - b - 1}{e^b}$.
You may assume that the condition for applying l'Hôpital's rule has been met.

```
AREA = (exp(b) - b - 1)/exp(b)
```

```
q3_2 = ...
```

```
verify_limit('3.2', q3_2, AREA, var=b,  
point=oo)
```

3.3 — May 2025 TZ2 Paper 2 Q11(c) (i), part of 5 marks

Use l'Hôpital's rule to find $\lim_{x \rightarrow \infty} (3x + 1)e^{-3x}$.

```
q3_3 = ...
```

```
verify_limit('3.3', q3_3, (3*x +  
1)*exp(-3*x), point=oo)
```

4. l'Hôpital's rule, more than one round

After differentiating, substitute again. Still $\frac{0}{0}$ — apply the rule again. Not indeterminate any more — stop, and read the answer. A round too many gives the wrong number and costs the mark.

4.1 — May 2024 TZ2 Paper 1 Q8, 6 marks

Use l'Hôpital's rule to find

$$\lim_{x \rightarrow 0} \frac{\sec^4 x - \cos^2 x}{x^4 - x^2}.$$

```
q4_1 = ...
```

```
verify_limit('4.1', q4_1, (sec(x)**4 -  
cos(x)**2)/(x**4 - x**2))
```

4.2 — May 2025 TZ3 Paper 1 Q9, 6 marks

Determine the value of $\lim_{x \rightarrow 0} \left(\frac{x \sin x}{1 - \cos x} \right)$.

```
q4_2 = ...
```

```
verify_limit('4.2', q4_2, x*sin(x)/(1 -  
cos(x)))
```

4.3 — May 2022 TZ2 Paper 2 Q7(b), 6 marks

Consider $\lim_{x \rightarrow 0} \frac{\arctan(\cos x) - k}{x^2}$, where $k \in \mathbb{R}$, and take $k = \frac{\pi}{4}$.

Using l'Hôpital's rule, show algebraically that the value of the limit is $-\frac{1}{4}$.

```
q4_3 = ...
```

```
verify_limit('4.3', q4_3, (atan(cos(x)) -  
pi/4)/x**2)
```

5. Maclaurin instead

Near zero a series *is* the function. Replace each factor by its expansion and read off the lowest power that survives. Where the denominator is x^3 or x^4 this is one line against three rounds of differentiation.

5.1 — November 2021 Paper 1 Q11(c), 4 marks

Let $f(x) = x^2 e^x$. Determine the value of

$$\lim_{x \rightarrow 0} \left[\frac{(x^2 e^x - x^2)^3}{x^9} \right].$$

```
q5_1 = ...
```

```
verify_limit('5.1', q5_1, (x**2*exp(x) -  
x**2)**3 / x**9)
```

5.2 — May 2022 TZ1 Paper 1 Q12(d) and Q12(e), 5 and 3 marks

The function g is defined by $g(x) = e^x \cos x$, where $x \in \mathbb{R}$.

(a) Find the Maclaurin series for $g(x)$ up to and including the x^4 term.

(b) Hence, or otherwise, determine the value of $\lim_{x \rightarrow 0} \frac{e^x \cos x - 1 - x}{x^3}$.

```
q5_2a = ...
```

```
q5_2b = ...
```

```
check_series('5.2a', q5_2a, '6a49f54210cb')  
verify_limit('5.2b', q5_2b, (exp(x)*cos(x) -  
1 - x)/x**3)
```

5.3 — May 2025 TZ1 Paper 1 Q12(e), 5 marks

Consider the family of functions $f_n(x) = \cos^n x$, where $x \in \mathbb{R}$ and $n \in \mathbb{N}$.

(a) Find the Maclaurin series of $f_n(x)$ up to the term in x^2 .

(b) Hence or otherwise, find $\lim_{x \rightarrow 0} \frac{f_n(x) - 1}{x^2}$ in terms of n .

```
q5_3a = ...
```

```
q5_3b = ...
```

```
check_series('5.3a', q5_3a, '281b04dc414d')  
verify_limit('5.3b', q5_3b, (cos(x)**n -  
1)/x**2, params={n: (1, 2, 5, 9)})
```

6. The constant that makes a limit exist

The letter is the answer. If the denominator goes to zero and the limit is finite, the numerator must go to zero too — and that equation gives the letter. Or the limit is stated and the letter is read out of it.

6.1 — May 2022 TZ2 Paper 2 Q7(a), 2 marks

Consider $\lim_{x \rightarrow 0} \frac{\arctan(\cos x) - k}{x^2}$, where $k \in \mathbb{R}$.

Show that a finite limit only exists for $k = \frac{\pi}{4}$, and state the value of k .

```
q6_1 = ...
```

```
verify_exact('6.1', q6_1, pi/4)
```

6.2 — November 2023 TZ1 Paper 1 Q9(b), 6 marks

Consider the function $f(x) = \frac{\sin^2(kx)}{x^2}$, where $x \neq 0$ and $k \in \mathbb{R}^+$.

Given that $\lim_{x \rightarrow 0} f(x) = 16$, find the value of k .

```
q6_2 = ...
```

```
check_num('6.2', q6_2, 1, '4b227777d4dd')
```

6.3 — May 2025 TZ2 Paper 2 Q11(c) (ii), part of 5 marks

The time T , in minutes, that a spinning top is in motion can be modelled by the probability density function

$$f(t) = \begin{cases} kte^{-3t}, & t \geq 0 \\ 0, & \text{otherwise,} \end{cases} \quad k \in \mathbb{Z}^+,$$

and $\int_0^a f(t) dt = \frac{k}{9} [1 - (3a + 1)e^{-3a}]$ for $a \in \mathbb{R}^+$.

Hence, by considering $\lim_{a \rightarrow \infty} \int_0^a f(t) dt$, find the value of k .

```
q6_3 = ...
```

```
check_num('6.3', q6_3, 1, '19581e27de7c')
```

7. The limit taken in a parameter

x is not sacred. When the arrow sits under n or under α , everything that is not the limiting variable stays exactly where it is, and the answer is an expression in the letters that stayed.

7.1 — May 2025 TZ3 Paper 3 Q1(d), 4 marks

Consider $f^n(x) = m^n x + c \left(\frac{1 - m^n}{1 - m} \right)$, where $-1 < m < 1$.

As $n \rightarrow \infty$, the family of graphs $y = f^n(x)$ approaches the graph of a straight line L

. Determine the equation of L , giving your answer in terms of c and m . Enter the right-hand side only.

```
FN = m**n*x + c*(1 - m**n)/(1 - m)

q7_1 = ...

verify_limit('7.1', q7_1, FN, var=n,
point=oo,
              params={m: (Rational(1, 2), -
Rational(3, 4)),
                      c: (5, -2), x: (3, 7)})
```

7.2 — November 2022 Paper 2 Q7, 6 marks

Consider the vectors $\mathbf{u} = \mathbf{i} + \mathbf{j}$ and $\mathbf{v} = \left(\cos \frac{1}{n}\right)\mathbf{i} + \left(\sin \frac{1}{n}\right)\mathbf{j}$, where $n \in \mathbb{Z}^+$. Let θ be the angle between $\mathbf{u}$ and $\mathbf{v}$.

(a) Find an expression for $\cos \theta$ in terms of n .

(b) Find the exact value of the limit approached by θ as $n \rightarrow \infty$.

```

q7_2a = ...
q7_2b = ...

verify_identity('7.2a', q7_2a, (cos(1/n) +
sin(1/n))/sqrt(2), var=n,
                samples=(1, 2, 3, 5))
check_num('7.2b', q7_2b, 6, '0e0202e1d463')

```

7.3 — November 2023 Paper 3 Q2(f), 2 marks

Two families of curves F and G intersect at an acute angle α . The gradient of F is $f(x, y)$, the gradient of G is $g(x, y)$, and

$$g(x, y) = \frac{f(x, y) + \tan \alpha}{1 - f(x, y) \tan \alpha}.$$

By considering $\lim_{\alpha \rightarrow \pi/2} \tan \alpha$, show that for all finite $f(x, y)$,

$$\lim_{\alpha \rightarrow \pi/2} g(x, y) = -\frac{1}{f(x, y)}.$$

Enter the limit as an expression in the letter **F**, standing for $f(x, y)$.

```
G = (F + tan(alpha))/(1 - F*tan(alpha))

q7_3 = ...

verify_limit('7.3', q7_3, G, var=alpha,
point=pi/2,
              params={F: (2, -3, Rational(1,
5))}))
```

8. The rule with a letter inside the expression

The limit is still in x , but a parameter sits inside: x^{n+2} , $S_n(x)$. Differentiate in x and carry the letter along; the answer comes out as a formula, or as one value that holds for every n at once. Before the rule comes the check, and here that check is worth its own mark.

8.1 — *November 2022 Paper 3 Q1(g)(i), 1 mark*

Let $f(x) = 1 + x + x^2 + \cdots + x^n$, $n \in \mathbb{Z}^+$, and $f_1(x) = xf'(x)$. For $x \neq 1$ it has been shown that

$$f_1(x) = \frac{nx^{n+2} - (n+1)x^{n+1} + x}{(x-1)^2}.$$

Show that $\lim_{x \rightarrow 1} f_1(x)$ is in indeterminate form, and name the form as a string: '0/0' or 'oo/oo'.

```
TOP = n*x**(n + 2) - (n + 1)*x**(n + 1) + x
F1 = TOP/(x - 1)**2
```

```
q8_1 = ...
```

```
verify_indeterminate('8.1', q8_1, TOP, (x - 1)**2, point=1,
                      params={n: (2, 5, 8)})
```

8.2 — November 2022 Paper 3 Q1(g)(ii), 5 marks

Hence, by applying l'Hôpital's rule, show that $\lim_{x \rightarrow 1} f_1(x) = \frac{1}{2}n(n+1)$.

```
q8_2 = ...
```

```
verify_limit('8.2', q8_2, F1, point=1,
             params={n: (2, 3, 7)})
```

8.3 — November 2025 TZ3 Paper 3 Q1(g), 3 marks

Let $S_1(x) = \sin x$ and $S_n(x) = \sin(S_{n-1}(x))$ for $n \geq 2$. It has been proved that

$$\begin{aligned} S'_n(x) &= \cos(S_{n-1}(x)) \cos(S_{n-2}(x)) \cdots \cos(S_1(x)) \cos x. \end{aligned}$$

Use l'Hôpital's rule to show that $\lim_{x \rightarrow 0} \frac{S_n(x)}{x} = 1$ for $n \in \mathbb{Z}^+$.

```
S3 = sin(sin(sin(x)))
S6 = sin(sin(sin(sin(sin(sin(x))))))

q8_3 = ...

verify_limit('8.3 (n = 3)', q8_3, S3/x)
verify_limit('8.3 (n = 6)', q8_3, S6/x)
```

9. What the limit means

Two marks in the topic are for a sentence about the thing being modelled. Talk about the runner and the parabola, not about the denominator.

9.1 — May 2025 TZ3 Paper 2 Q10(c) (ii), part of 3 marks

With $v(t) = \frac{8.14t}{\sqrt{t^2 + 0.2}}$ as in 2.2, state a reason why the value in 2.2 is not valid in the context of this question.

This question has no cell: the answer is a sentence. Write it below and compare with the solution.

your sentence here

**9.2 — May 2025 TZ1 Paper 3 Q2(b)
(iii), 2 marks**

The curvature of a function f is $k(x) = \frac{|f''(x)|}{(1 + (f'(x))^2)^{3/2}}$, and for a quadratic $h(x) = ax^2 + bx + c$ with $a \neq 0$ it is given that

$$k(x) = \frac{2|a|}{(1 + (2ax + b)^2)^{3/2}}.$$

State the value of $\lim_{x \rightarrow \infty} k(x)$ and explain briefly the significance of this result. The cell checks the value; write the explanation in the markdown cell below it.

```
KX = 2*abs(a) / (1 + (2*a*x +  
b)**2)**Rational(3, 2)  
  
q9_2 = ...  
  
verify_limit('9.2', q9_2, KX, point=oo,  
params={a: (2, -3), b: (5, 0)})
```

your explanation here



Solutions

Numbered as above. Read one after you have worked the question.

1.1 — May 2024 TZ1 P1 Q11(f)(i)

$P(-1) = -3 + 5 - 1 - 1 = 0$, so $(x + 1)$ divides P , and $P(x) = (x + 1)(3x^2 + 2x - 1) = (x + 1)^2(3x - 1)$. With $Q(x) = (x + 1)(2x + 1)$,

$$f(x) = \frac{(x + 1)^2(3x - 1)}{(x + 1)^2(2x + 1)} = \frac{3x - 1}{2x + 1} \quad (x \neq -1),$$

$$\text{so } \lim_{x \rightarrow -1} f(x) = \frac{-4}{-1} = 4.$$

No calculus is needed. The cancelled $(x + 1)^2$ is the shared reason both parts vanish; once it is gone the function is continuous at -1 and substitution finishes the job. Note that $f(-1)$ itself does not exist — which is the whole reason the question asks for a limit.

2.1 — May 2024 TZ1 P1 Q11(f)(ii)

Using the cancelled form and dividing by x ,

$$\lim_{x \rightarrow \infty} \frac{3x - 1}{2x + 1} = \lim_{x \rightarrow \infty} \frac{3 - \frac{1}{x}}{2 + \frac{1}{x}} = \frac{3}{2}.$$

Top and bottom have the same degree, so the limit is the ratio of the leading coefficients.

2.2 — May 2025 TZ3 P2 Q10(c)(i)

$$v(t) = \frac{8.14t}{\sqrt{t^2 + 0.2}} = \frac{8.14}{\sqrt{1 + \frac{0.2}{t^2}}} \rightarrow 8.14.$$

Write down means one mark and no working expected — but this is the working that tells you the answer is 8.14 and not 0 or ∞ .

3.1 — May 2021 TZ1 P1 Q8, 5 marks

Substituting gives $\frac{\arctan 0}{\tan 0} = \frac{0}{0}$, so the rule applies:

$$\lim_{x \rightarrow 0} \frac{\arctan 2x}{\tan 3x} = \lim_{x \rightarrow 0} \frac{\frac{2}{1+4x^2}}{3 \sec^2 3x} = \frac{2}{3}.$$

M1 for attempting to differentiate numerator and denominator, **A1A1** for the two derivatives, **(M1)** for substituting $x = 0$, **A1** for $\frac{2}{3}$. The markscheme note: *do not condone absence of limits* — award a maximum of **M1A1A0M1A1** if lim is missing.

3.2 — May 2023 TZ1 P3 Q1(c)(i), 2 marks

The form is $\frac{\infty}{\infty}$:

$$\lim_{b \rightarrow \infty} \frac{e^b - b - 1}{e^b} = \lim_{b \rightarrow \infty} \frac{e^b - 1}{e^b} = \lim_{b \rightarrow \infty} \frac{e^b}{e^b} = 1.$$

A1 for the correct quotient — *condone absence of limit* here — and **A1** for the 1. In the question this is A_1 , the whole area under xe^{-x} from 0 to infinity, and it is exactly 1.

3.3 — May 2025 TZ2 P2 Q11(c)(i)

$(3x + 1)e^{-3x}$ is a product of the form $\infty \cdot 0$, and the rule does not apply to products. Rewrite first:

$$\begin{aligned}\lim_{x \rightarrow \infty} (3x + 1)e^{-3x} &= \lim_{x \rightarrow \infty} \frac{3x + 1}{e^{3x}} \\ &= \lim_{x \rightarrow \infty} \frac{3}{3e^{3x}} \\ &= \lim_{x \rightarrow \infty} \frac{1}{e^{3x}} \\ &= 0.\end{aligned}$$

The markscheme's note on the first line is unusually firm: *this first **A1** must be seen*. The rewrite is the mark.

4.1 — May 2024 TZ2 P1 Q8, 6 marks

$\frac{0}{0}$ at the start. First round:

$$\lim_{x \rightarrow 0} \frac{4 \sec^4 x \tan x + 2 \sin x \cos x}{4x^3 - 2x},$$

still $\frac{0}{0}$. Second round:

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{16 \sec^4 x \tan^2 x + 4 \sec^6 x - 2 \sin^2 x + 2 \cos^2 x}{12x^2 - 2} &= \frac{0 + 4 - 0 + 2}{-2} \\ &= -3.\end{aligned}$$

The denominator is now -2 , so the form is no longer indeterminate and the work

stops. **M1** for the second application providing their expression is in indeterminate form as $x \rightarrow 0$ and providing there is no third attempt at using l'Hôpital's rule, and: to award full marks limit notation $\lim_{x \rightarrow 0}$ must be seen at least once; if no limit notation is seen but otherwise all correct, do not award the final **A1**.

4.2 — May 2025 TZ3 P1 Q9, 6 marks

$\frac{0 \cdot 0}{1-1} = \frac{0}{0}$. First round: $\lim_{x \rightarrow 0} \frac{\sin x + x \cos x}{\sin x}$, still $\frac{0}{0}$. Second round:

$$\lim_{x \rightarrow 0} \frac{2 \cos x - x \sin x}{\cos x} = \frac{2 - 0}{1} = 2.$$

Without any calculus: multiply above and below by $1 + \cos x$,

$$\frac{x \sin x (1 + \cos x)}{1 - \cos^2 x} = \frac{x}{\sin x} (1 + \cos x) \rightarrow 1 \cdot 2 = 2.$$

4.3 — May 2022 TZ2 P2 Q7(b), 6 marks

With $k = \frac{\pi}{4}$ the form is $\frac{0}{0}$. Differentiating,

$$\frac{d}{dx} \arctan(\cos x) = \frac{-\sin x}{1 + \cos^2 x},$$

so

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{-\sin x}{2x(1 + \cos^2 x)} &= -\frac{1}{2} \cdot \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \lim_{x \rightarrow 0} \frac{1}{1 + \cos^2 x} \\ &= -\frac{1}{2} \cdot 1 \cdot \frac{1}{2} \\ &= -\frac{1}{4}. \end{aligned}$$

Or apply the rule a second time to the same effect. Since the answer is printed in the question, every mark is for the working.

5.1 — November 2021 P1 Q11(c), 4 marks

$x^9 = (x^3)^3$, so pull the cube out first:

$$\begin{aligned} \frac{(x^2 e^x - x^2)^3}{x^9} &= \left(\frac{x^2(e^x - 1)}{x^3} \right)^3 \\ &= \left(\frac{e^x - 1}{x} \right)^3. \end{aligned}$$

And $e^x - 1 = x + \frac{x^2}{2} + \dots$, so $\frac{e^x - 1}{x} \rightarrow 1$ and the limit is $1^3 = 1$. The markscheme prints this as METHOD 2 and applies l'Hôpital's rule to the inner quotient; METHOD 1 uses the Maclaurin series from the previous part.

5.2 — May 2022 TZ1 P1 Q12(d) and Q12(e)

(a) With $g(x) = e^x \cos x$: $g(0) = 1$, $g'(0) = 1$, $g''(0) = 0$, $g'''(0) = -2$, $g^{(4)}(0) = -4$, so

$$g(x) = 1 + x - \frac{x^3}{3} - \frac{x^4}{6} + \dots$$

The vanishing x^2 term is what makes the next part work.

(b)

$$\frac{e^x \cos x - 1 - x}{x^3} = \frac{-\frac{x^3}{3} - \frac{x^4}{6} + \dots}{x^3} \rightarrow -\frac{1}{3}.$$

The l'Hôpital route needs three rounds and runs to half a page; the markscheme prints both.

5.3 — May 2025 TZ1 P1 Q12(e), 5 marks

(a) $\cos x = 1 - \frac{x^2}{2} + \dots$, so

$$\cos^n x = \left(1 - \frac{x^2}{2} + \dots\right)^n = 1 - \frac{nx^2}{2} + \dots$$

— only the first two binomial terms matter, because the rest carry x^4 . Equivalently $f_n(0) = 1$, $f'_n(0) = 0$, $f''_n(0) = -n$.

(b)

$$\frac{f_n(x) - 1}{x^2} = \frac{-\frac{nx^2}{2} + \dots}{x^2} \longrightarrow -\frac{n}{2}.$$

Markscheme note: *do not award FT marks for an expression that does not involve n* . An answer without n in it is not an answer.

6.1 — May 2022 TZ2 P2 Q7(a), 2 marks

$\lim_{x \rightarrow 0} x^2 = 0$, so the indeterminate form $\frac{0}{0}$ is *required* for the limit to be finite — otherwise the quotient is unbounded. Hence

$$\lim_{x \rightarrow 0} (\arctan(\cos x) - k) = 0$$

$$\implies \arctan 1 - k$$

$$= 0$$

$$\implies k$$

$$= \frac{\pi}{4}.$$

M1 for setting the numerator's limit to zero, **A1** for $k = \arctan 1$, **AG** for $\frac{\pi}{4}$. The note: *award **M1A0** for using $k = \frac{\pi}{4}$ to show the limit is $\frac{0}{0}$ — that is the argument backwards.*

6.2 — November 2023 TZ1 P1 Q9(b), 6 marks

$$\frac{\sin^2 kx}{x^2} = \left(\frac{\sin kx}{x} \right)^2 = k^2 \left(\frac{\sin kx}{kx} \right)^2 \longrightarrow k^2,$$

or apply l'Hôpital's rule twice to the same end. So $k^2 = 16$, and $k \in \mathbb{R}^+$ leaves $k = 4$. Rejecting -4 out loud is part of the answer.

6.3 — May 2025 TZ2 P2 Q11(c)(ii)

Take the limit through the given integral, using 3.3:

$$\begin{aligned}\lim_{a \rightarrow \infty} \int_0^a f(t) dt &= \frac{k}{9} \left(1 - \lim_{a \rightarrow \infty} (3a + 1)e^{-3a} \right) \\ &= \frac{k}{9}.\end{aligned}$$

A probability density function integrates to 1 over its whole range — **M1**, *recognising area of probability density function = 1, seen anywhere* — so $\frac{k}{9} = 1$ and $k = 9$.

7.1 — May 2025 TZ3 P3 Q1(d), 4 marks

Only m^n moves, and $-1 < m < 1$ gives $m^n \rightarrow 0$. So $m^n x \rightarrow 0$ and $c \frac{1 - m^n}{1 - m} \rightarrow \frac{c}{1 - m}$, leaving

$$L : y = \frac{c}{1 - m}.$$

(M1) for applying $m^n \rightarrow 0$ to at least one term, **(A1)** for the term in x vanishing, **A1** for the constant term, **A1** for the line.

Note that x never moved: L is horizontal, which is the point of the question.

7.2 — November 2022 P2 Q7, 6 marks

(a) $\mathbf{u} \cdot \mathbf{v} = \cos \frac{1}{n} + \sin \frac{1}{n}$, $|\mathbf{u}| = \sqrt{2}$ and $|\mathbf{v}| = 1$, so

$$\cos \theta = \frac{\cos \frac{1}{n} + \sin \frac{1}{n}}{\sqrt{2}}.$$

(b) As $n \rightarrow \infty$, $\frac{1}{n} \rightarrow 0$, so $\cos \theta \rightarrow \frac{1+0}{\sqrt{2}} = \frac{1}{\sqrt{2}}$ and $\theta \rightarrow \frac{\pi}{4}$. Equivalently $\mathbf{v} \rightarrow \mathbf{i}$, and the angle between $\mathbf{i} + \mathbf{j}$ and $\mathbf{i}$ is $\frac{\pi}{4}$. The note: *accept 45° ; do not accept rounded values such as 0.785.*

7.3 — November 2023 P3 Q2(f), 2 marks

As $\alpha \rightarrow \frac{\pi}{2}$, $\tan \alpha \rightarrow \infty$, so $\frac{1}{\tan \alpha} \rightarrow 0$. Divide the top and bottom of g by $\tan \alpha$:

$$g = \frac{\frac{f}{\tan \alpha} + 1}{\frac{1}{\tan \alpha} - f} \longrightarrow \frac{0 + 1}{0 - f} = -\frac{1}{f}.$$

M1 for the division, **R1** for $\frac{1}{\tan \alpha} \rightarrow 0$, and the **R1** is dependent on the **M1**. Geometrically: $\alpha = \frac{\pi}{2}$ means the families are perpendicular, and perpendicular gradients are negative reciprocals.

8.1 — November 2022 P3 Q1(g)(i), 1 mark

At $x = 1$ the numerator is $n \cdot 1 - (n + 1) \cdot 1 + 1 = 0$ and the denominator is $(1 - 1)^2 = 0$, so

$$\lim_{x \rightarrow 1} f_1(x) = \frac{n - (n + 1) + 1}{0} = \frac{0}{0}.$$

R1, and the note is strict: *only award **R1** for sufficient simplification of the numerator, as shown above. Do not award **R1** if $\lim_{x \rightarrow 1}$ is not referred to or stated.*

8.2 — November 2022 P3 Q1(g)(ii), 5 marks

First round:

$$\lim_{x \rightarrow 1} \frac{n(n+2)x^{n+1} - (n+1)^2x^n + 1}{2(x-1)}.$$

At $x = 1$ the numerator is $n(n+2) - (n+1)^2 + 1 = 0$, so it is $\frac{0}{0}$ again. Second round:

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{n(n+1)(n+2)x^n - n(n+1)^2x^{n-1}}{2} \\ = \frac{n(n+1)[(n+2) - (n+1)]}{2} = \frac{n(n+1)}{2}. \end{aligned}$$

Which is $1 + 2 + \cdots + n$ — as it had to be, since $f_1(1) = 1 + 2 + \cdots + n$ directly from the definition.

8.3 — November 2025 TZ3 P3 Q1(g), 3 marks

$S_n(0) = 0$ and the denominator is 0, so the form is $\frac{0}{0}$. One round of the rule:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{S_n(x)}{x} &= \lim_{x \rightarrow 0} \frac{S'_n(x)}{1} \\ &= \lim_{x \rightarrow 0} \cos(S_{n-1}(x)) \cdots \cos(S_1(x)) \cos x. \end{aligned}$$

Every $S_k(0) = 0$, so every factor tends to $\cos 0 = 1$, and a product of n ones is 1. The induction in the previous part is what makes this three lines rather than n of them.

9.1 — May 2025 TZ3 P2 Q10(c)(ii)

The race is only 200 metres long. Fiona finishes it in about 26 seconds, so t never approaches infinity and her velocity never reaches 8.14 m s^{-1} : the limit describes the formula after the race has stopped describing the runner.

Answers about the algebra — "the denominator grows without bound" — earn nothing here. The mark is for talking about the race.

9.2 — May 2025 TZ1 P3 Q2(b)(iii), 2 marks

As $x \rightarrow \infty$ the term $(2ax + b)^2$ grows without bound while the numerator $2|a|$ is fixed, so

$$\lim_{x \rightarrow \infty} k(x) = 0.$$

A1. And the significance, **R1:** the curvature of a quadratic tends to zero far from the vertex — *for large positive values of x , quadratic functions are close*

to being straight, or, as the note allows, a quadratic function behaves like a linear function.

Where the marks are

§	technique	questions	marks
1	Substitute and cancel	1	part of 3
2	Highest power at infinity	2	part of 6
3	l'Hôpital, one round	3	8
4	l'Hôpital, more than one	3	18
5	Maclaurin instead	3	9
6	The constant that makes it exist	3	10
7	The limit in a parameter	3	9
8	The rule with a letter inside	3	9
9	What the limit means	2	part of 5

Sections 3 and 4 together are 26 of the 73 marks and they are one instruction repeated: substitute, name the form,

differentiate, substitute again. Sections 7 and 8 are the surprise of the topic — eighteen marks where the answer is a formula in a letter rather than a number — and that is where a student who has only practised $\frac{0}{0}$ loses the most.